

$$\varepsilon_0 = 8,85 \times 10^{-12} C^2 N^{-1} m^{-2}$$

$$\mu_0 = 4\pi \times 10^{-7} Tm/A$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

$$\mathcal{E} = -\frac{d}{dt}\Phi_B$$

$$\mathcal{E}_N = -N \frac{d}{dt}\Phi_B$$

$$\mathcal{E} = BLv$$

$$L = N \frac{\Phi_B}{i}$$

$$L = \mu_0 N^2 \frac{A}{l}$$

$$\mathcal{E} = -L \frac{d}{dt}i$$

$$U = \frac{1}{2} LI^2$$

$$\tau = L/R$$

$$i(t) = \frac{V}{R}(1 - e^{-t/\tau})$$

$$V_L(t) = V e^{-t/\tau}$$

$$i(t) = \frac{V}{R} e^{-t/\tau}$$

$$V_L(t) = -V e^{-t/\tau}$$

$$\frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{1}{LC} q = 0$$

$$\gamma = R/2L$$

$$\omega_0 = 1/\sqrt{LC}$$

$$U_E = \frac{1}{2C} q^2(t)$$

$$U_B = \frac{1}{2} Li^2(t)$$

$$E = \frac{Q^2}{2C} = \frac{L\omega_0^2 Q^2}{2}$$

$$i(t) = I \cos(\omega t)$$

$$I_{rms} = I/\sqrt{2}$$

$$V_{rms} = V/\sqrt{2}$$

$$v_R(t) = V_R \cos(\omega t)$$

$$\bar{P} = \frac{1}{2} V_R I$$

$$v_L(t) = V_L \cos(\omega t + \pi/2)$$

$$X_L = \omega L$$

$$V_L = IX_L$$

$$v_C(t) = V_C \cos(\omega t - \pi/2)$$

$$X_C = 1/\omega C$$

$$V_C = IX_C$$

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$V = IZ$$

$$v(t) = V \cos(\omega t + \phi)$$

$$\tan \phi = \frac{X_L - X_C}{R}$$

$$\bar{P} = \frac{1}{2} VI \cos \phi$$

$$= V_{rms} I_{rms} \cos \phi$$

$$V_{rms} = Z I_{rms}$$

$$\frac{V_P}{N_P} = \frac{V_S}{N_S}$$

$$I_P V_P = I_S V_S$$

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{1}{\varepsilon_0} Q_{int}$$

$$\oint_S \vec{B} \cdot d\vec{A} = 0$$

$$\oint_C \vec{E} \cdot d\vec{s} = -\frac{d}{dt} \oint_S \vec{B} \cdot d\vec{A}$$

$$\oint_C \vec{B} \cdot d\vec{s} = \mu_0 I + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

$$\nabla \cdot \vec{E} = \frac{1}{\varepsilon_0} \rho$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$$

$$\nabla^2 \vec{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{B} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

$$\frac{\partial^2 \vec{E}_y}{\partial x^2} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}_y}{\partial t^2}$$

$$\frac{\partial^2 \vec{B}_z}{\partial x^2} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}_z}{\partial t^2}$$

$$E_y(x, t) = E_0 \cos(kx - \omega t)$$

$$B_z(x, t) = B_0 \cos(kx - \omega t)$$

$$\frac{\omega}{k} = c = \frac{E_0}{B_0}$$

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

$$S = \frac{E_0 B_0}{\mu_0} \sin^2(kx - \omega t)$$

$$I_{em} = \frac{1}{2} \frac{E_0 B_0}{\mu_0}$$

$$I_{em} = \bar{S} = c \bar{\eta}_{em}$$

$$\bar{\eta}_{em} = \frac{1}{2} \varepsilon_0 E_0^2$$